MTH636 – Game Theory

Lecture 25: Kuhn's Theorem & Equilibria in Behavior Strategies

1 Recap: Constructing an Equivalent Behavior Strategy from a Mixed Strategy

From the previous lecture, when we have a mixed strategy σ_1 for Player I and want to find an equivalent behavior strategy b_1, we use the following approach via a worked example.

1.1 Example: Two-Level Game Tree

Consider a game tree where Player I has two information sets: $h_1 = \{n_0\}$ (the root, actions L with prob. u and R with prob. $1-u$) and $h_2 = \{n_1, n_2\}$ (actions ℓ and r). Player I's pure strategies are $L\ell$, Lr, $R\ell$, Rr.

Suppose the mixed strategy satisfies:

$$\sigma_1(L\ell) = \sigma_1(Rr) = \tfrac{1}{2}.$$

The terminal nodes have labels as shown; for Player II the behavior strategy at h_2 uses probabilities y and $(1-y)$ for ℓ and r:

$$\sigma_1(L\ell) = \tfrac{1}{2}, \quad \sigma_1(Rr) = \tfrac{1}{2}.$$

Let $b_1(L; n_0) = u$ and $b_1(\ell; h_2) = y$. Then the reaching probabilities give:

$$uy = \tfrac{1}{2}, \quad u(1-y) = 0, \quad (1-u)y = 0, \quad (1-u)(1-y) = \tfrac{1}{2}.$$

This system has the solution: $u = \frac{1}{2}$, $y = 1$ is *not* consistent; instead the only solution is $u = 1, y = 1/2$ fails too. In fact the compatible solution is $u = 1/2$, which implies from $u(1-y) = 0$ that $y = 1$, contradiction with $uy = 1/2$ giving $y = 1$. Re-examining: $uy = 1/2$ and $(1-u)(1-y) = 1/2$, and $u(1-y) = 0$ so $u = 0$ or $y = 1$. Since $u = 0$ contradicts $uy = 1/2$, we need $y = 1$. But then $(1-u)(1-y) = 0 \neq 1/2$. Hence **no solution exists** for this imperfect-recall tree.

Now consider a different (perfect-recall) tree. Suppose:

$$\sigma_1(AL) = \tfrac{1}{5}, \quad \sigma_1(Ar) = \tfrac{1}{10}, \quad \sigma_1(BL) = \tfrac{2}{5}, \quad \sigma_1(Br) = \tfrac{3}{10}.$$

Here $\sigma_1(A\cdot) = \frac{1}{5} + \frac{1}{10} = \frac{3}{10}$ and $\sigma_1(B\cdot) = \frac{2}{5} + \frac{3}{10} = \frac{7}{10}$... (see boxed equations below).

For the tree with n_0 (Player I, actions A prob. u and B prob. $1-u$), n_1 (Player II, actions L and R), and an information set $h_2 = \{n_3, n_4\}$ (Player I, actions ℓ prob. y and r prob. $1-y$):

$$\boxed{ny = \tfrac{1}{5}, \quad n(1-y) = \tfrac{1}{10}, \quad (1-n)y = \tfrac{2}{5}, \quad (1-n)(1-y) = \tfrac{3}{10}.}$$

Solving: $n(y + 1 - y) = ny + n(1-y) = \frac{1}{5} + \frac{1}{10} = \frac{3}{10}$, so $n = \frac{3}{10}$... Checking sign: from $ny = \frac{1}{5}$ with $n = 3/10$: $y = \frac{1/5}{3/10} = \frac{2}{3}$. Similarly $(1-n) = 7/10$.

2 Constructing b_i from σ_i – General Formula

Definition: Converting a Mixed Strategy to a Behavior Strategy

Let $S_i(h_2) \subseteq S_i$ denote the set of strategies of Player i such that the game **possibly reaches** h_2 (i.e., the information set h_2 is reachable under s_i).

More precisely, for information set h_2, define:

$$S_i(\ell;\, h_2) \subseteq S_i(h_2)$$

as the set of strategies in $S_i(h_2)$ that **choose action** ℓ **at** h_2.

Also define:

$$\sigma_i\Big(S_i(h_2)\Big) : \text{ the total probability of } S_i(h_2) \text{ under } \sigma_i,$$
$$\sigma_i\Big(S_i(\ell;\, h_2)\Big) : \text{ the total probability of } S_i(\ell;\, h_2) \text{ under } \sigma_i.$$

The behavior strategy b_i is then constructed as:

$$b_i(\ell;\, h_2) = \begin{cases} \dfrac{\sigma_i\Big(S_i(\ell;\, h_2)\Big)}{\sigma_i\Big(S_i(h_2)\Big)} & \text{provided } \sigma_i\Big(S_i(h_2)\Big) > 0, \\ \alpha \in [0,1] & \text{o.f. (otherwise free)}, \end{cases}$$

and similarly:

$$b_i(r;\, h_2) = \begin{cases} \dfrac{\sigma_i\Big(S_i(r;\, h_2)\Big)}{\sigma_i\Big(S_i(h_2)\Big)} & \text{provided } \sigma_i\Big(S_i(h_2)\Big) > 0, \\ 1-\alpha & \text{o.f.} \end{cases}$$

2.1 Computed Example

For the information set $h_1 = S_i$ (the root):

$$S_i(h_1) = S_i.$$

$$S_i(A;\, h_1) = \{A\ell,\, Ar\}.$$

$$b_i(A;\, h_1) = \frac{\sigma_i\Big(S_i(A;\, h_1)\Big)}{\sigma_i(S_i)} = \frac{\sigma_i(A\ell) + \sigma_i(Ar)}{1} = \tfrac{1}{5} + \tfrac{1}{10} = \tfrac{3}{10}.$$

Note

When $\sigma_i(S_i(h_2)) = 0$ (the information set is reached with zero probability under σ_i), the value of $b_i(\ell;\, h_2)$ can be chosen *freely* (any $\alpha \in [0,1]$). This is why the conversion from mixed strategy to behavior strategy is generally non-unique only when some information sets are off the equilibrium path.

3 Kuhn's Theorem (1957)

Theorem: Kuhn's Theorem (1957)

In every game in extensive form, if Player i has **perfect recall**, then for every mixed strategy of Player i, there exists a behavior strategy that is **equivalent** to the mixed strategy.

Note

Together with the result from Lecture 24 (that given any behavior strategy one can always construct an equivalent mixed strategy), Kuhn's Theorem establishes that **under perfect recall**, the sets of mixed strategies and behavior strategies are *strategically equivalent* – every mixed strategy has an equivalent behavior strategy and vice versa.

4 Equilibria in Behavior Strategies

Theorem: Existence of NE in Behavior Strategies

If all players in an extensive form game have **perfect recall**, then the game has a Nash Equilibrium in behavior strategies.

Proof. From the extensive form game, construct the corresponding strategic (normal) form game.

Since the extensive form game is finite, the corresponding strategic form game is finite too.

Now, the strategic form game has a mixed strategy Nash Equilibrium – say $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$. For each σ_i^*, construct b_i^*, an outcome-equivalent behavior strategy (which exists by Kuhn's Theorem, since each player has perfect recall). We claim $b^* = (b_1^*, \ldots, b_n^*)$ is an NE in behavior strategies.

By Remark R-2 from Lecture 24:

$$U_i(\sigma^*) = U_i(b^*) \quad \forall\, i \in N.$$

Assume for contradiction that b^* is **not** an NE. Then $\exists\, k \in N$ such that:

$$U_k(b^*) < U_k(b_k,\, b_{-k}^*) \quad \text{for some behavior strategy } b_k.$$

Now, since b_k is a behavior strategy, there exists an equivalent mixed strategy σ_k (this direction always holds). Therefore:

$$U_k(b^*) < U_k(b_k,\, b_{-k}^*) = U_k(\sigma_k,\, b_{-k}^*).$$

Since σ^* is equivalent to b^* component-wise and σ_k is equivalent to b_k:

$$= U_k(\sigma_k,\, \sigma_{-k}^*) \leq U_k(\sigma^*), \quad \text{a contradiction.}$$

(The last inequality holds because σ^* is a Nash Equilibrium of the strategic form game, so no player can profitably deviate.)

Hence b^* is an NE in behavior strategies. □

Key Takeaway

Key Takeaway. For extensive form games with perfect recall:

1. Every mixed strategy $\Leftrightarrow$ some behavior strategy (Kuhn's Theorem, plus the easy direction).
2. A Nash Equilibrium in *mixed* strategies always induces a Nash Equilibrium in *behavior* strategies.
3. Behavior strategies are preferred in practice because they are far more compact to describe.